

Viyom Shukla

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GitHub: github.com/viyomshukla

TUF+: 350+ DSA problems | LeetCode: 200+ problems | GFG: 100+ problems

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Portfolio: viyom-potfolio.vercel.app

Location: Noida, India

SKILLS SUMMARY

- **Languages:** C++ (Proficient), Python, JavaScript, C, SQL
- **Frontend:** React.js, Tailwind CSS, HTML/CSS, Chrome Extensions
- **Backend & APIs:** Node.js, Express.js, REST APIs, Google OAuth 2.0, JWT Authentication, Razorpay, Twilio, Gemini/Replicate APIs
- **Databases, Cloud:** MongoDB Atlas, MySQL, Firebase, AWS (EC2, S3), Cloudinary, Git, GitHub
- **Core CS:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Distributed & Scalable Systems, OOP, Machine Learning

EDUCATION

- **Jaypee Institute of Information Technology** Noida, India
Integrated B.Tech + M.Tech in Computer Science Engineering; CGPA: 7.62/10 2022 – 2027
- **Gayatri Public School** Agra, India
Class XII (CBSE): 88.6% | Class X (CBSE): 86.6%

EXPERIENCE

- **Distance Connect** Remote
Web Developer Intern Aug 2025 – Oct 2025
 - **Platform UI & Architecture:** : Spearheaded frontend development for an early-stage EdTech startup connecting students with IIT and AIIMS mentors, comprehensively revamping the user interface to enhance student engagement.
 - **AI Chatbot Integration:** : Engineered and deployed an AI-powered conversational interface to automate student onboarding and rapidly resolve queries, significantly reducing manual support overhead.
 - **Real-Time Mentorship Pipeline:** : Architected a seamless video consultation flow by integrating the Google Meet API, enabling secure 1:1 virtual mentoring sessions directly within the platform.

PROJECTS

- **Thumblify — AI-Powered Thumbnail Generation SaaS (MERN, Distributed Systems)** *Source Code*
 - **AI Generation Engine:** Architected a full-stack SaaS serving 100+ users; generated 3 distinct thumbnail styles in 2K/4K via Gemini and Replicate APIs with fault-tolerant processing.
 - **Secure Authentication:** Implemented Google OAuth 2.0 sign-in with JWT session tokens, safeguarding all user accounts and API endpoints.
 - **Payments & Monetization:** Launched a credit-based subscription model via Razorpay, securing 100% of transactions with signature verification and webhooks.
 - **Conversational Interfaces:** Created 2 conversational channels — an in-app AI chatbot and a WhatsApp bot via Twilio API — enabling single-prompt generation from mobile.
 - **In-Browser Editing:** Embedded TUI Image Editor with 10+ editing tools (crop, filters, text overlays), eliminating reliance on external design software.
 - **Performance & Delivery:** Delivered 1,000+ generated images via Cloudinary CDN and shipped a Chrome extension backed by a low-latency backend.
- **DeepMood — Depression Prediction Using GANs (Machine Learning)** *Notebook*
 - **Synthetic Data Pipeline:** Devised a Generative Adversarial Network (GAN) pipeline that synthesized 5,000+ mental-health records, mitigating class imbalance in clinical datasets.
 - **Model Development:** Executed end-to-end preprocessing and exploratory analysis; benchmarked 4 classifiers on real and GAN-augmented data, achieving 92% accuracy.
 - **Evaluation & Validation:** Evaluated models across 4 metrics (precision, recall, F1-score, confusion matrix), minimizing false negatives in clinical screening.
- **NextTrip — Tour Booking & Management Platform (MERN)** *Source Code*
 - **Role-Based Platform:** Engineered a booking platform with Google OAuth 2.0 login and JWT role-based access for 3 roles (admin, organizer, traveler), securing 100% of API routes.
 - **Organizer & Identity Module:** Empowered organizers to publish trips with rich image galleries; built an Aadhaar document-upload flow with admin review and manual approval before verifying travelers for KYC-compliant onboarding.
 - **Payments:** Automated booking payments via Razorpay API with server-side verification of each transaction, reducing manual reconciliation to 0.
 - **Admin Control & Data Layer:** Built an admin dashboard for trip listings, promotional imagery, and booking analytics on a normalized 6-collection MongoDB schema with indexed queries.

ACHIEVEMENTS

- **Research Publications — 2 Papers on IEEE (IC3, IIIT Noida):**
 - (1) *Enhancing Network Intrusion Detection Using Conditional GANs for Imbalanced Data Classification* — [IEEE Xplore](#): achieved 99.8% accuracy on NSL-KDD by synthesizing minority attack-class data with Conditional GANs and ensemble methods (Random Forest, SVM, Neural Networks).
 - (2) *Enhancing Network Intrusion Detection Using Quantum GANs for Imbalanced Data Classification* — accepted at IC3-2026 (IEEE Xplore, forthcoming): achieved 99.9% accuracy on NSL-KDD via hybrid quantum learning for rare attack samples.
- **International Conference Speaker:** Presented published research on Network Intrusion Detection at the 17th IEEE IC3-2025, IIIT Noida. [Certificate](#)
- **Competitive Programming:** Solved 350+ DSA problems on TUF+, 200+ DSA and 50+ SQL problems on LeetCode, and 100+ on GeeksforGeeks, focusing on optimization, time complexity, and algorithm design.
- **National Hackathons:** Competed in Smart India Hackathon 2023, Adobe India Hackathon '25, Flipkart Grid 6.0, and Amazon ML Challenge.
- **Student Placement Committee Member:** Guided 100+ students through their job search; coordinated placement drives and tracked companies, job roles, and hiring outcomes.